

MATH550 Numerical PDEs Homework #0

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Problem 1: Finite differences

1. Exact analytic solution $f'(x)$ and my numerical solution $f'_n(x)$:

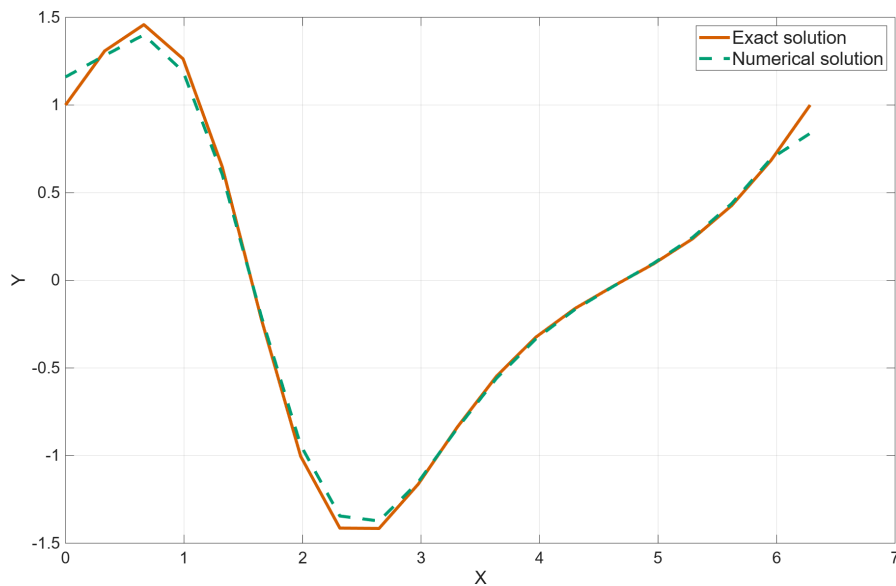


Figure 1: Chart of exact vs. numerical solution

2. Relative error convergence plot:

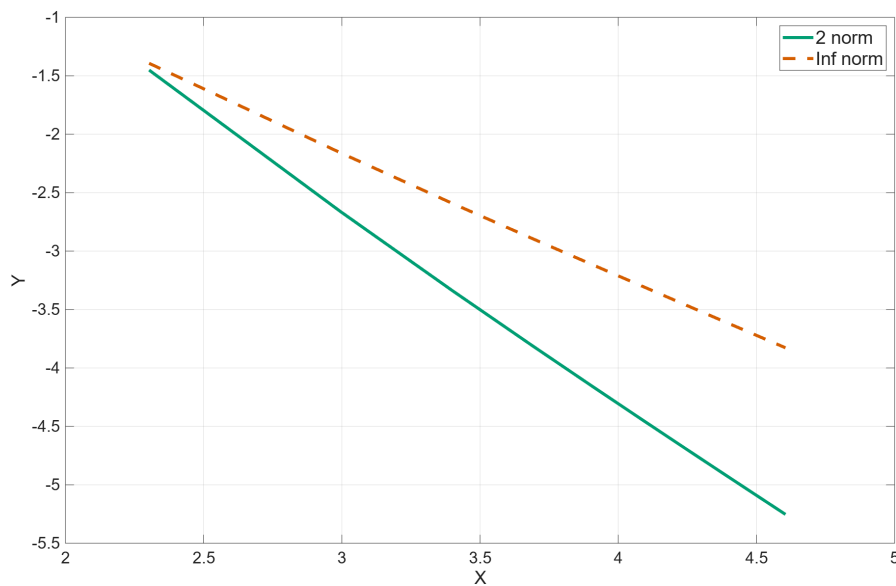


Figure 2: 2-Norm vs. Inf norm

Problem 2: Second-order derivatives

1. Exact analytic solution $f''(x)$ and my numerical solution $f''_n(x)$:

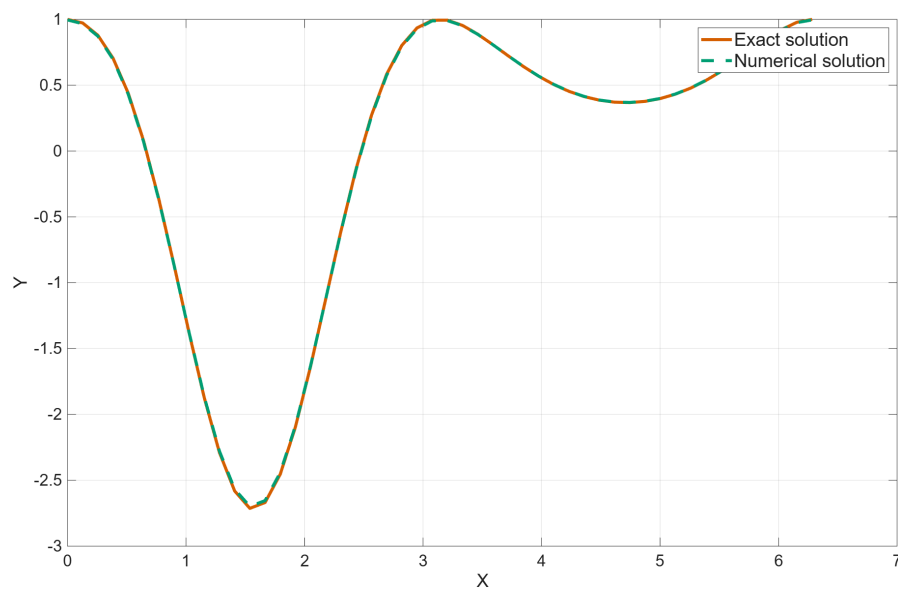


Figure 3: Chart of exact vs. numerical solution

2. Relative error convergence plot:

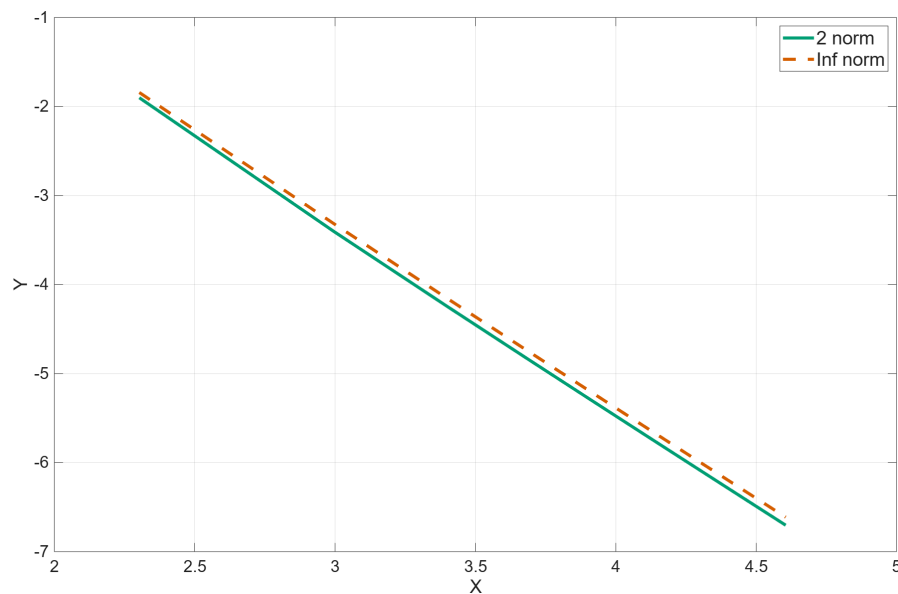


Figure 4: 2-Norm vs. Inf norm

Problem 3: ODE

$$\frac{d^2u}{dx^2} \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2}, \quad \frac{du}{dx} \approx \frac{u_{i+1} - u_{i-1}}{2\Delta x}$$

Plugging these values into the ODE, we get:

$$(a) \quad (2 + \sin(x)\Delta x)u_{i+1} + (2\Delta x^2 - 4)u_i + (2 - \sin(x)\Delta x)u_{i-1} \approx 2\Delta x^2 f(x)$$

If $u(x) = \sin(x)$ then we can solve for $f(x)$ directly:

$$\frac{d^2u}{dx^2} = -\sin(x), \quad \frac{du}{dx} = \cos(x) \Rightarrow f(x) = \sin(x)\cos(x)$$

We can also calculate the boundary conditions at 0 and 5 to solve for the function $u(x)$. The error plot of the numerical estimate (a) against the real solution is shown below:

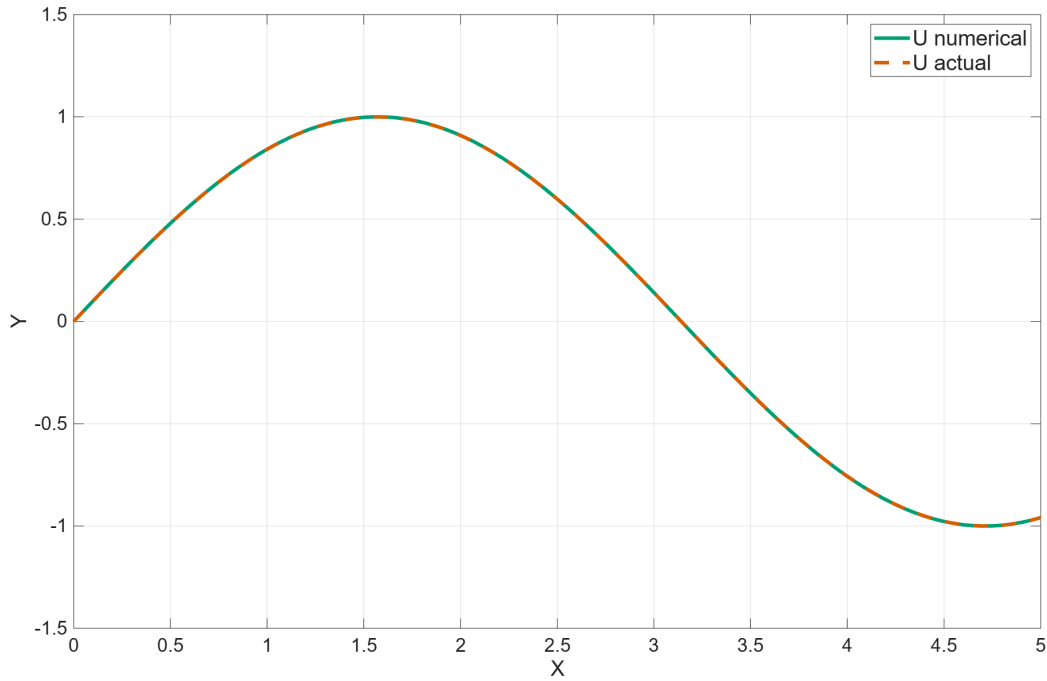


Figure 5: Chart of exact vs approximate solution

The error plot proves the method converges with second order:

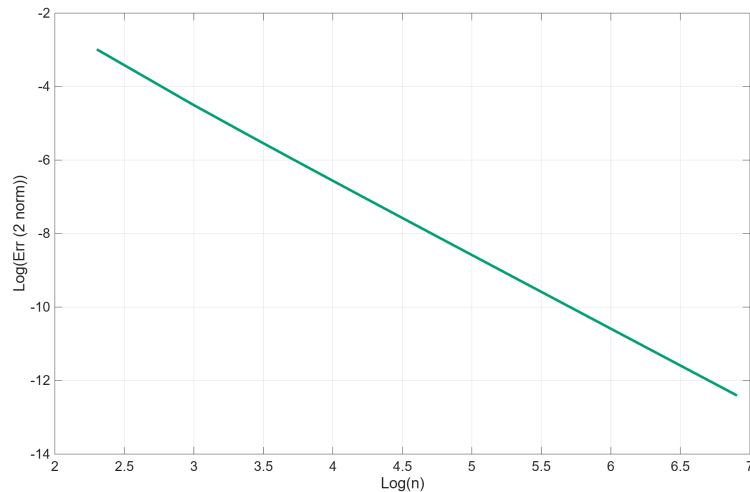


Figure 6: Chart of 2-norm error vs log(n grid points)

Problem 4: Poisson Equation

1.

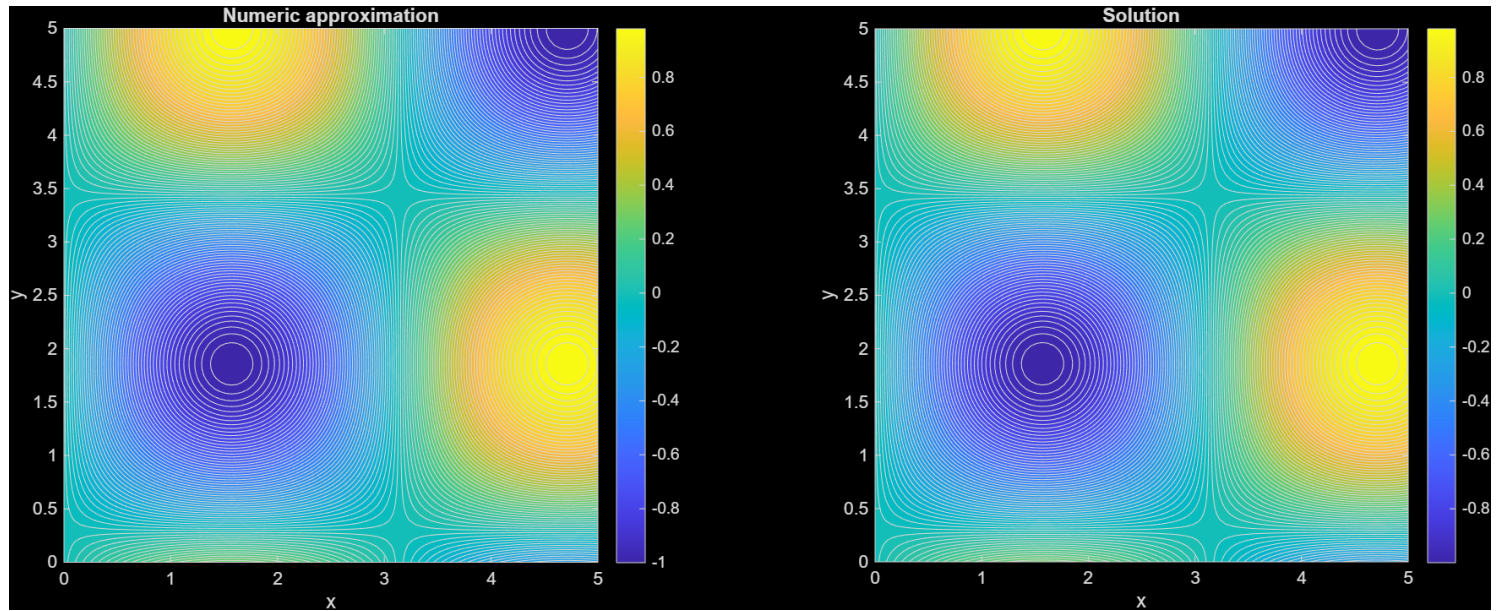


Figure 7: Chart of approximation vs solution

Use of AI

Models used: 90% Deepseek, 10% Copilot in MATLAB

In P1 I used MATLAB copilot to construct the finite difference matrix using `spdiags` and asked it different questions about MATLAB syntax and for the plot specifications.

In P2 I used deepseek for MATLAB syntax questions.

For P3 I sent deepseek the problem statement of solving an ODE using second-order finite difference methods with the Dirichlet conditions, along with information that I had been given u and needed to find the necessary forcing function f . I wasn't clear on what the question meant and at first thought I was estimating f , not u . Deepseek made it clear that I needed to estimate u instead, using boundary conditions given by the testing function u . I did not share any of the equations with it.

For P4 I used deepseek for MATLAB syntax questions.

I used deepseek for LaTeX syntax questions as well.